Molecular Techniques for Investigating Genetic Diseases

(Data Interpretation session)

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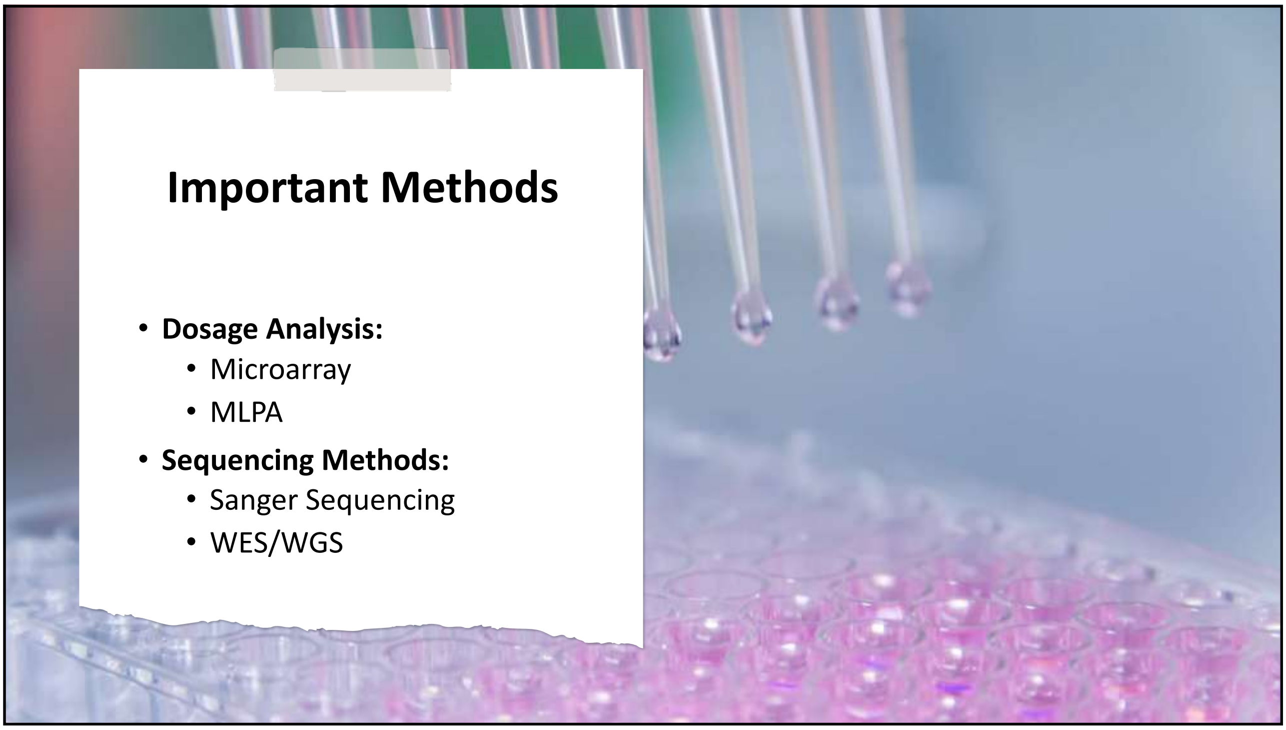
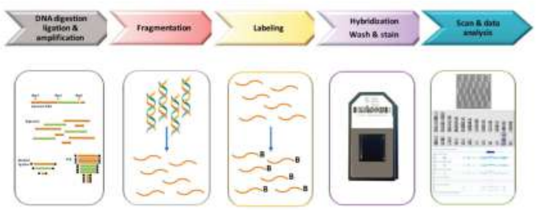


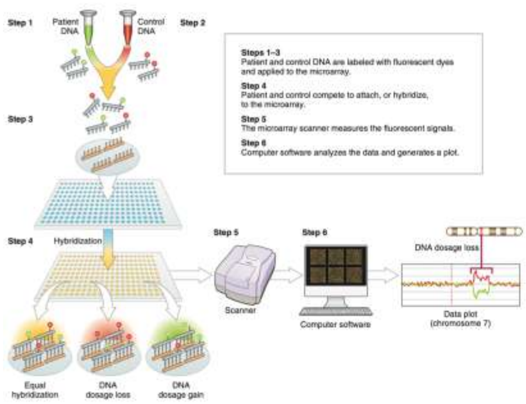
Important Methods

- **Dosage Analysis:**
  - Microarray
  - MLPA
- **Sequencing Methods:**
  - Sanger Sequencing
  - WES/WGS

Dosage Analysis:

- Microarray
- MLPA





Important Microarray Types

- SNP array: Single Nucleotide Polymorphism Array
- Array CGH: Array Comparative Genomic Hybridization

SNP array: Single Nucleotide Polymorphism

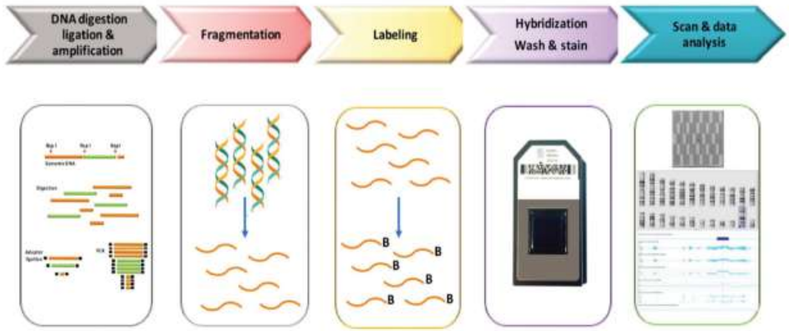


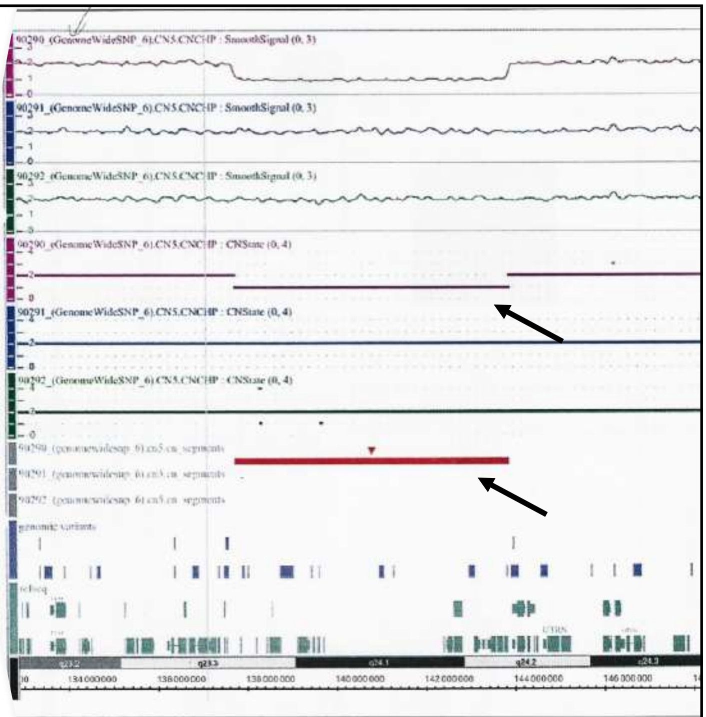
Fig. 2.4: Principles of CMA.

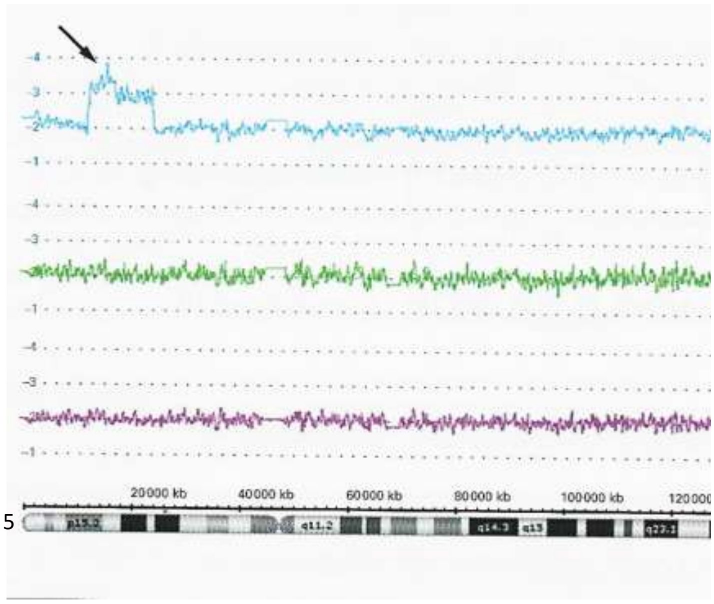
Belhassan, K., & Granadillo, J. L. (2021). Current approaches to genetic testing in pediatric disease. In *Biochemical and Molecular Basis of Pediatric Disease* (pp. 15-36).

1) Interpret the following result

- Name examples of diseases that can be diagnosed with this technique

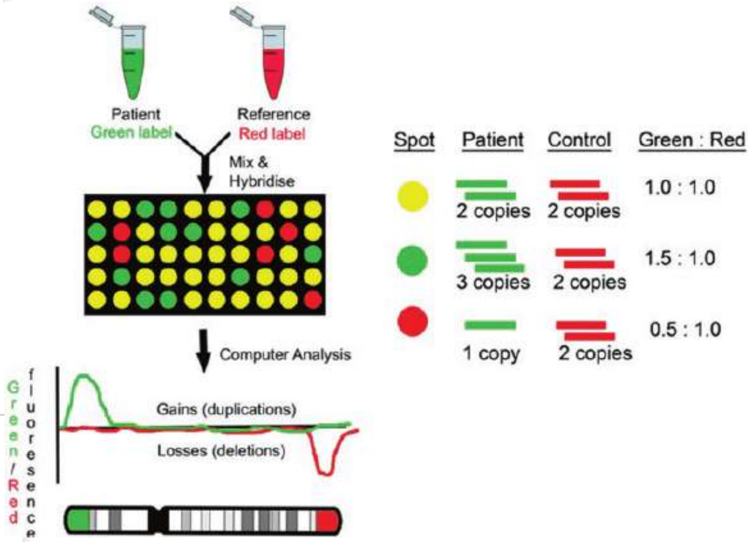
Chr. 6





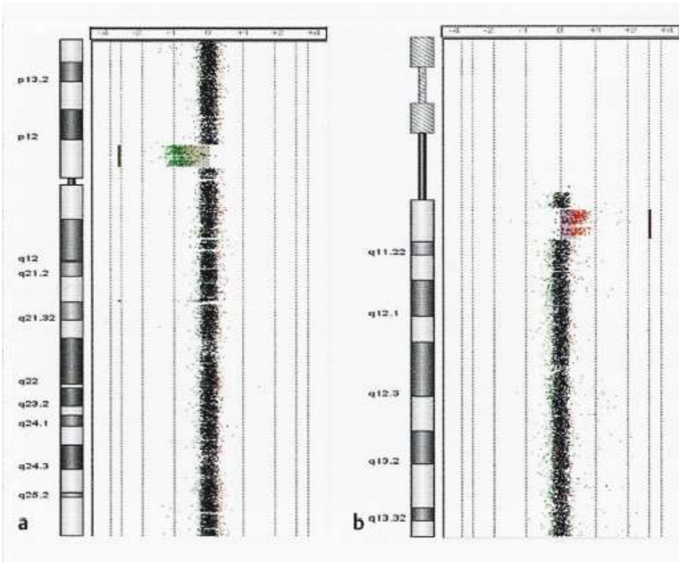
2) Interpret the following result

Array CGH: Comparative Genomic Hybridization

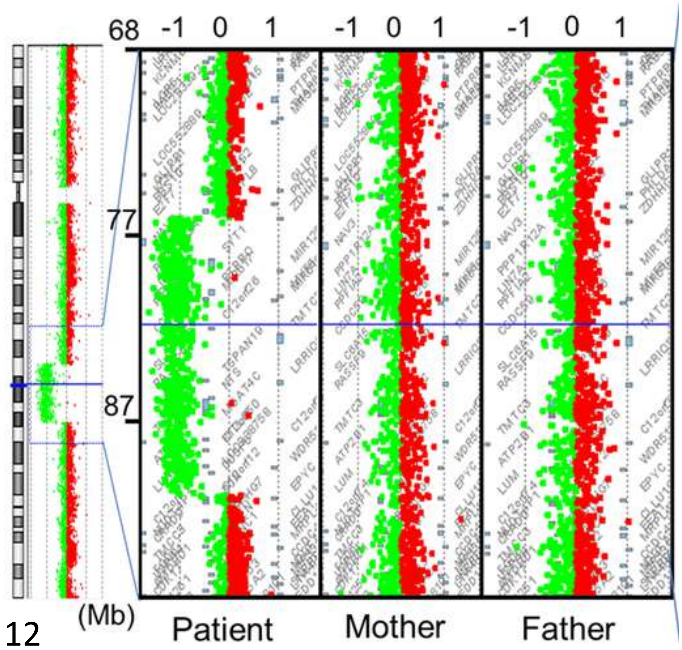


reference: https://www.researchgate.net/publication/313545452_Techniques_of_Chromosomal_Studies

1) Interpret the following results



2) Interpret the following results



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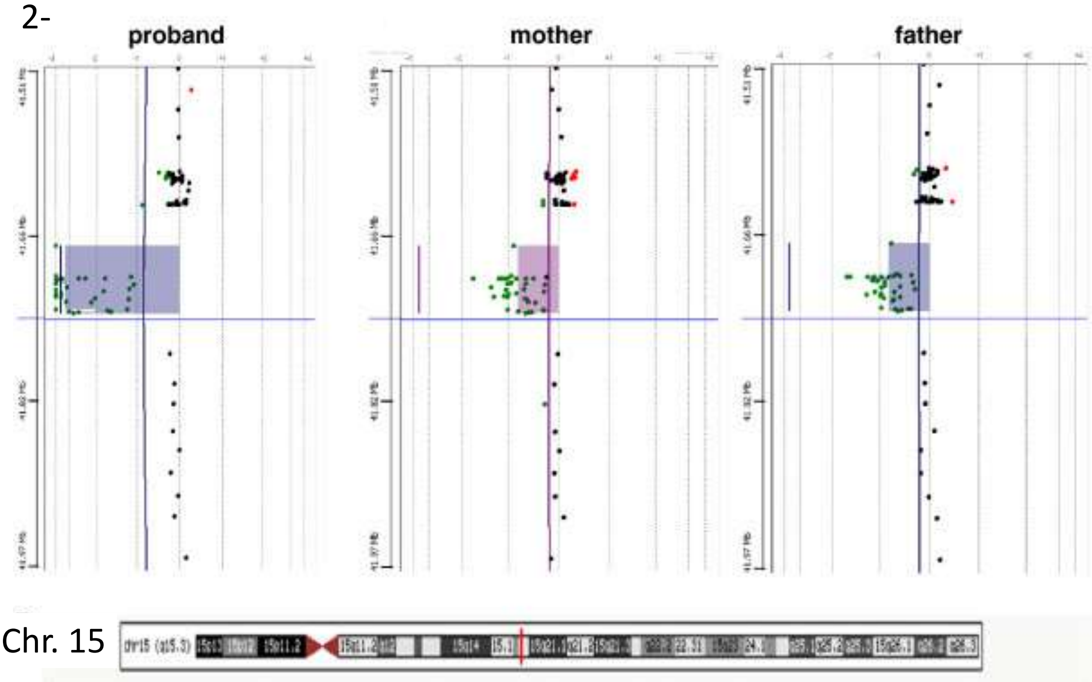
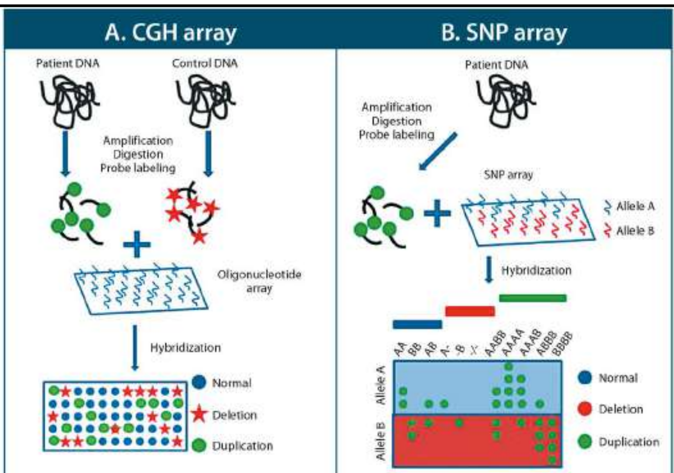




The comparative genomic hybridization (CGH) array compares the patient's DNA to control DNA using 2 different fluorescent labels. Labeled control and patient DNA fragments are hybridized to an array containing oligonucleotide DNA sequences from genes throughout the human genome. Each position on the array correlates to a different part of the genome. The relative intensity of the 2 different labels indicates copy-number changes. When only the red label (control DNA) is present, it indicates an absence of patient DNA and therefore a deletion (red stars). When there is more patient than control DNA, the patient label is overrepresented (green circles) and indicates duplication. When there are no copy-number changes, there should be equal amounts of control-labeled and patient-labeled DNA (indicated with blue circles).

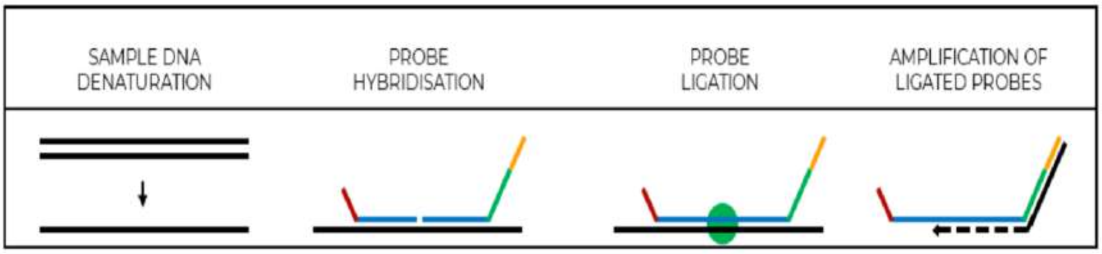
A single-nucleotide polymorphism (SNP) array contains small fragments of DNA from the human genome where there are known to be multiple alleles. Each allele is represented on the array and each position on the array corresponds to a genetic locus. DNA from the patient is hybridized to the array. Patients who have the A allele at a specific locus will bind to the A allele on the array. If the patient is homozygous, the sample will bind only to A or B (AA or BB). If the patient is a heterozygote, the sample will be label hybridized to A and B (AB). Copy-number changes are determined by the relative intensity of bound DNA at each allele with a relative decrease in deletions (red bar) and an increase in duplications (green bar). Consanguinity is indicated by a loss of heterozygosity over large spans of DNA.

FIGURE 1

Overview of chromosomal microarrays.

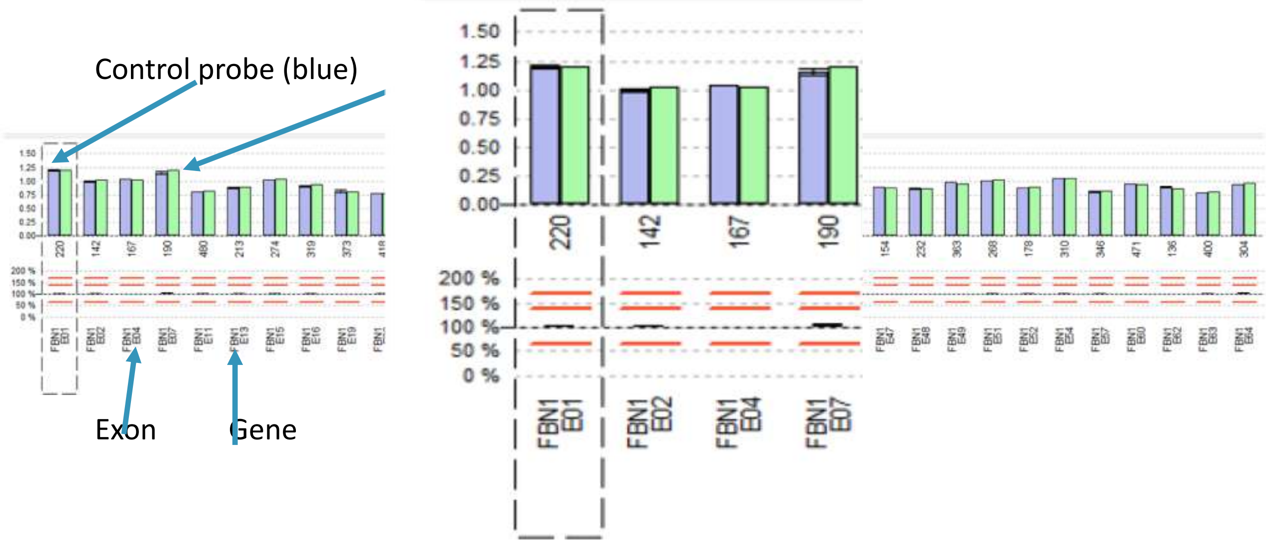
MLPA
(multiplex ligation-dependent probe amplification)

Methodological Principle of MLPA

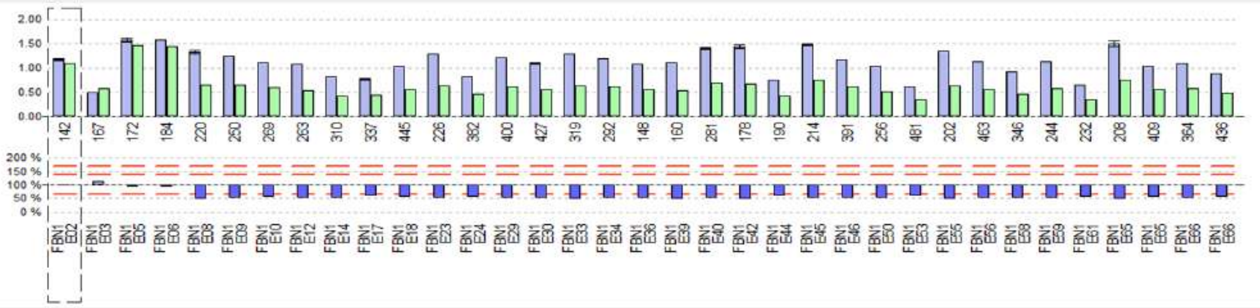


MLPA reaction

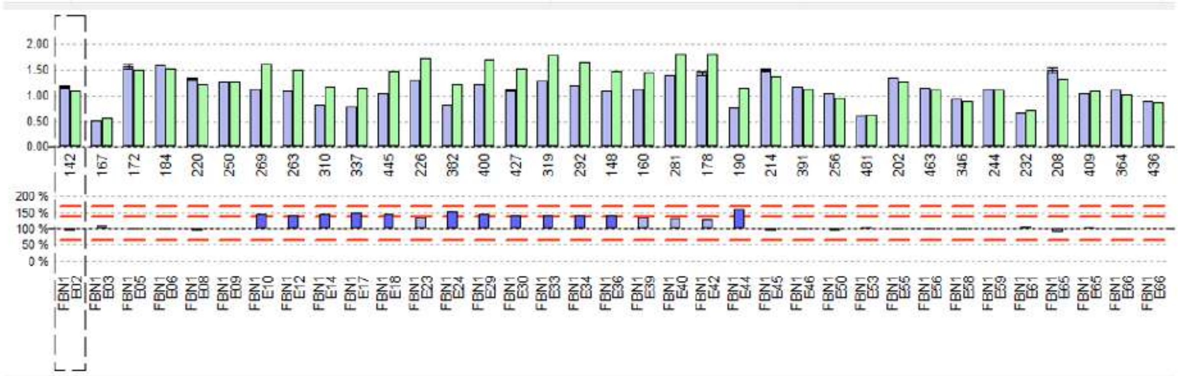
Examples of MLPA: Evaluation (Normal)



1- Interpret the following result:

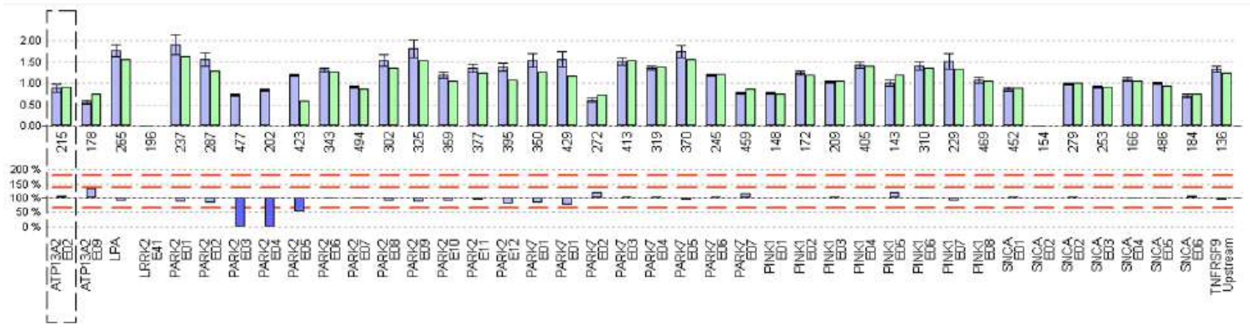


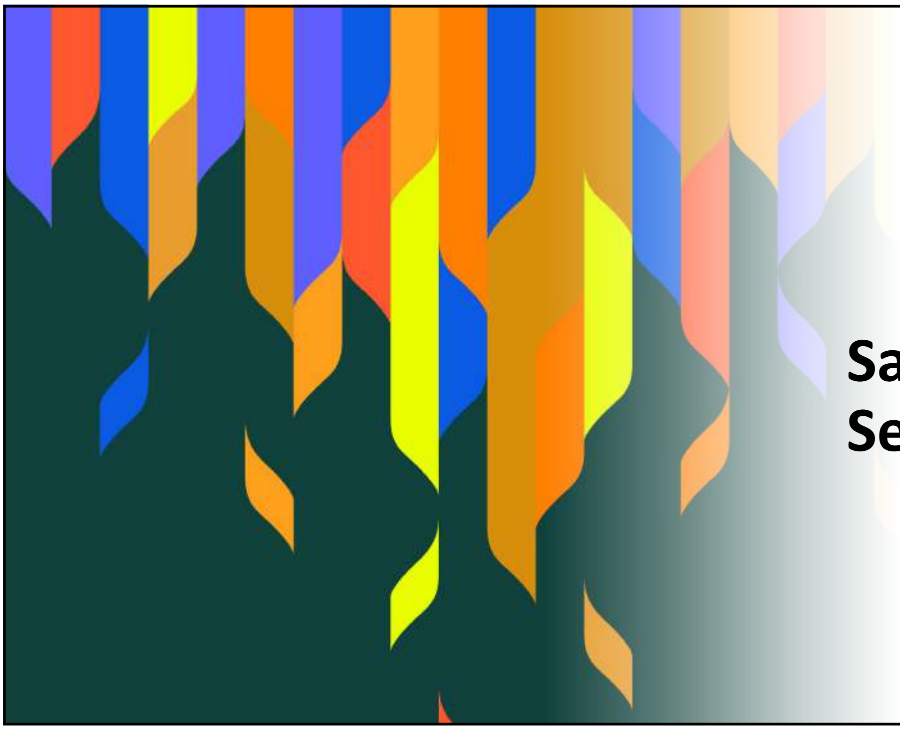
2- Interpret the following result:



3- Interpret the following result:

- Give examples of diseases that could be diagnosed with this technique

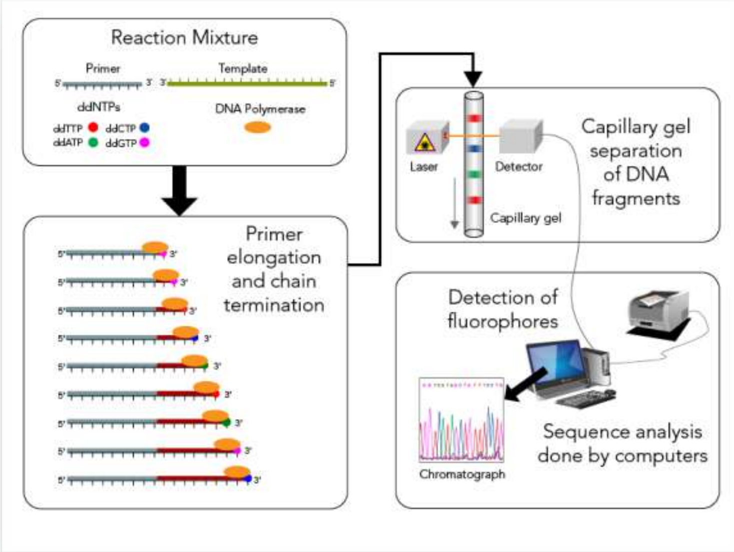




Sanger Sequencing

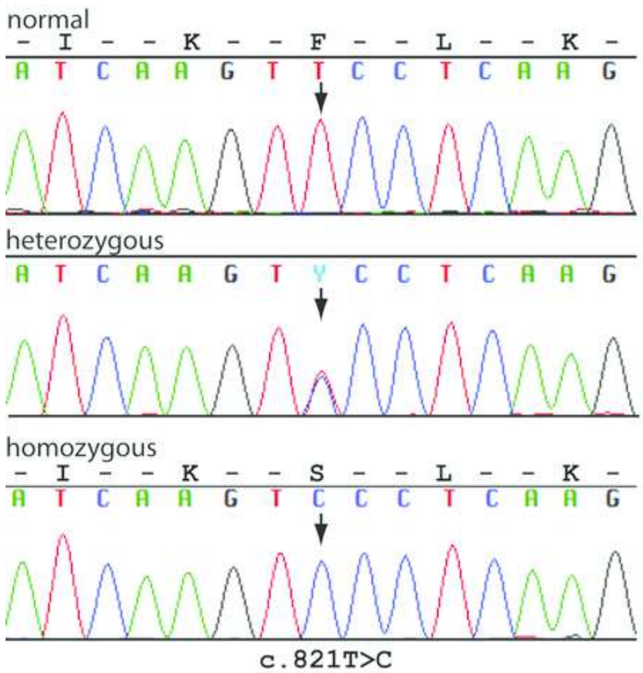
Necessary Components

- PCR-Product
- Primer
- DNA-Polymerase
- Free nucleotides
- Didesoxynucleotide (ddNTPs)
- Puffer solution



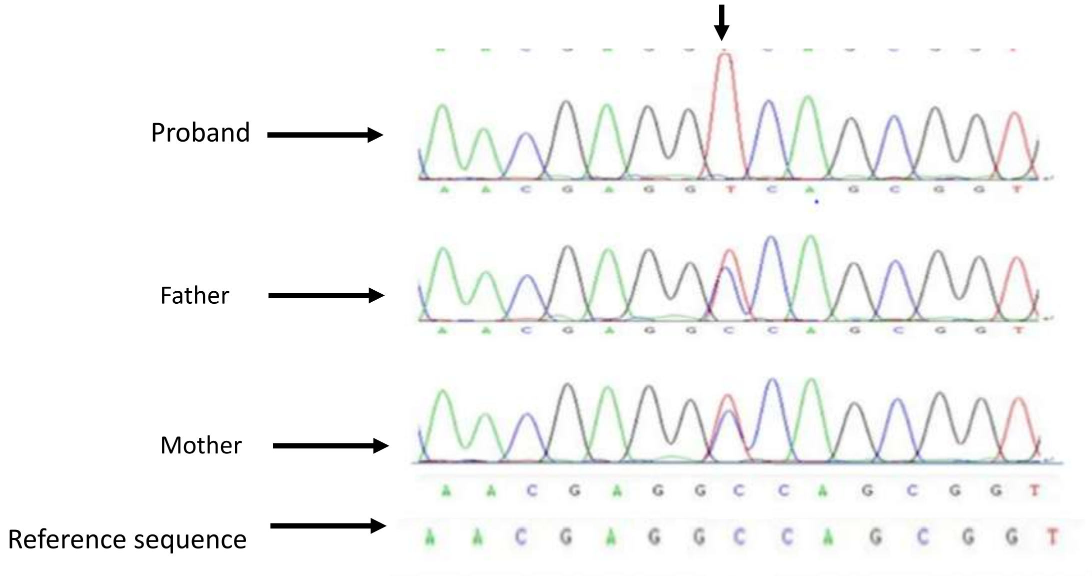
https://letstalkscience.ca/educational-resources/backgrounders/sanger-sequencing

Sanger Sequencing Interpretation

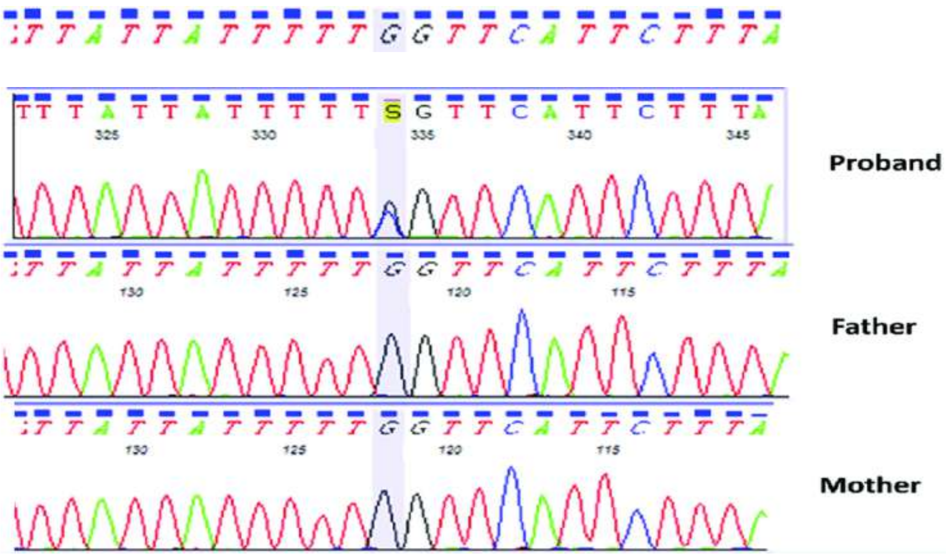


https://bio.davidson.edu/Courses/genomics/2016/Dwyer/assignment_1.html

1) Interpret the following results



Interpret the following results:



Next-Generation Sequencing (NGS)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	#ch	start	end	r	a	ge	sn	dep	m	variant	vari	re	gene	vari	coding	sn	snp	ljb_j	ljb_r	esp6	re	omim	quality
44	chr13	23911234	23911234	G	T	het	225	240	37	SNV	0.4549	exoni	SACS	nonsyr	SACS:NM_014363:exon10:c.C6781A:p.L2261I,	0.004	rs1467;	0.9997	0.8457	0.0052	.		604490	passed
154	chr6	16327865	16327870	TGC	-	het	217	882	44	INDEL	0.0014	exoni	ATXN1	nonfra	ATXN1:NM_000332:exon8:c.672_677del:p.224_226de	.	.	.	.	.	753	601556	passed
162	chr6	152473168	152473168	G	A	het	225	1335	37	SNV	0.5046	exoni	SYNE1	stopga	SYNE1:NM_182961:exon134:c.C24238T:p.Q8080X,SYN	.	.	0.9997	10.000	.	.	608441	passed
163	chr6	152501416	152501416	C	T	het	225	637	37	SNV	0.3826	exoni	SYNE1	nonsyr	SYNE1:NM_182961:exon129:c.G23315A:p.R7772Q,SY	0.002	rs13871	0.9716	0.9808	0.0055	.		608441	passed
165	chr6	152644651	152644652	GA	-	het	217	1253	37	INDEL	0.0027	exoni	SYNE1	frames	SYNE1:NM_182961:exon82:c.15878_15879del:p.5293	.	.	.	.	.	.	608441	passed

201
202
203 SYNE1:NM_182961:exon134:c.C24238T:p.Q8080X, validieren; VUS4; 23.04.2013 PB validiert 01.07.2013 JKB
204 SYNE1:NM_182961:exon82:c.15878_15879del; validieren; VUS4; 23.04.2013 PB validiert 11.07.13 JKB
205
206

	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD
1	vari	re	gene	vari	coding	sn	snp	ljb_j	ljb_r	esp6	re	omim	quality	ihdb	ihdb	ihdb	ih	v	(	dated
44	0.4549	exoni	SACS	nonsyr	SACS:NM_014363:exon10:c.C6781A:p.L2261I,	0.004	rs1467;	0.9997	0.8457	0.0052	.	604490	passed	0.0000	0.0000	10.000	0	.	.	
154	0.0014	exoni	ATXN1	nonfra	ATXN1:NM_000332:exon8:c.672_677del:p.224_226de	.	.	.	.	.	753	601556	passed	0.0000	0.0000	10.000	0	.	.	
162	0.5046	exoni	SYNE1	stopga	SYNE1:NM_182961:exon134:c.C24238T:p.Q8080X,SYN	.	.	0.9997	10.000	.	.	608441	passed	0.0000	0.0000	10.000	0	.	.	
163	0.3826	exoni	SYNE1	nonsyr	SYNE1:NM_182961:exon129:c.G23315A:p.R7772Q,SY	0.002	rs13871	0.9716	0.9808	0.0055	.	608441	passed	0.0000	0.0000	10.000	0	.	.	
165	0.0027	exoni	SYNE1	frames	SYNE1:NM_182961:exon82:c.15878_15879del:p.5293	.	.	.	.	.	.	608441	passed	0.0000	0.0000	10.000	0	.	.	

201
202
203 _182961:exon134:c.C24238T:p.Q8080X, validieren; VUS4; 23.04.2013 PB validiert 01.07.2013 JKB
204 _182961:exon82:c.15878_15879del; validieren; VUS4; 23.04.2013 PB validiert 11.07.13 JKB
205
206

Example of evaluation and analysis with NGS technique

QUESTIONS?DISCUSSION!

THANK YOU